Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

Database Management Systems Practicum

ICT 1222

Assignment 01 – Mini Project

Group 14

Submitted to: Ms.Sandaruwani Pathirage

Submitted by:
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1. Introduction to the Problem

Faculty of Technology (FOT), University of Ruhuna handles hundreds of students, many departments, and many courses each semester. Before implementing this application there was not an efficient way of handling the academic management process and resulted in many important problems.

- Attendance management process was done separately for each lecture/course without any method of obtaining attendance percentage for each student or alerting lecturers when a particular student falls under 80 percent attendance requirement for examinations.
- The process of calculating continuous assessment marks from quizzes, projects, mid-exam was done manually and prone to errors.
- No method existed for automatically calculating whether a student is eligible to appear for final exams based on CA marks and attendance.
- GPA calculation for final exams was a lengthy process that needed manual calculation of weighted average of marks in all courses taken by the particular student.
- No role-based access control and all types of users (Dean, Lecturer, Technical Officer, Students) could use the application with equal permissions.
- Students could apply for medical leave but without any systematic method of updating their attendance.
- For repeat students who take the same course twice, there was no system for management of this problem.

This is what led to the creation of the FOT Academic Management System, which is a completely normalized, relational database management system based on MySQL that solves all the above problems in an organized way..

2. Introduction to the Solution

The FOT Academic Management System database is a relational database management system implemented on MySQL database. The implementation has been done within four functional modules that are highly integrated:

- **Users Management** – controls users in the database through generalization hierarchy starting from PERSON, then specialized to STUDENT and STAFF, where STAFF itself gets specialized further into ADMIN, DEAN, LECTURER, and TECHNICAL_OFFICER. A person can have a USER_ACCOUNT associated with his or her record for logging into the system.

- **Academic Management** – includes DEPARTMENT, SEMESTER, COURSE_UNIT, COURSE_OFFERING (which is an offering of a particular course unit for a specific batch in a semester), COURSE_COMPONENT (Theory/Practical), and ENROLLMENT (M:N relationship between STUDENTS and COURSE_OFFERINGs).

- **Attendance Management** – consists of SESSION information for each component, ATTENDANCE_RECORD for each individual session (for example, present/absent/medical), and MEDICAL_RECORD for the medically approved absence.

- **Marks & Results Management** – holds ASSESSMENT_SCHEME (marks scheme for each course_offering), STUDENT_MARK (raw marks), ELIGIBILITY_RECORD (attendance percentage and CA marks), FINAL_RESULT (final mark, grade, and result code), GPA_RECORD (SGPA and CGPA), and GRADE_SCALE (mark to grade conversion table).

Referential integrity is enforced by foreign keys, data quality is maintained using constraints such as NOT NULL, UNIQUE, and ENUM, derived data is revealed using 15 database views, and all the business rules are automatically handled through three stored procedures. MySQL users have role-based accounts that adhere to the least privilege principle.

3. Proposed ER / EER Diagram

ER/EER diagrams were created using draw.io, and it captures all the entities, attributes, relationships, cardinalities, participation constraints, and specialization hierarchy of the FOT Academic Management system. Below are some of the decisions made during the designing of the diagram.

3.1 Generalisation / Specialisation Hierarchy

The supertype entity is PERSON and it includes the following attributes: Person_ID, F_Name, L_Name, NIC, Gender, DOB, Phone_no, and Address. The PERSON entity is then specialized into STUDENT and STAFF disjointly and partially. Additionally, STAFF is specialized disjointly and totally into ADMIN, DEAN, LECTURER, and TECHNICAL OFFICER entities.

3.2 Academic Core

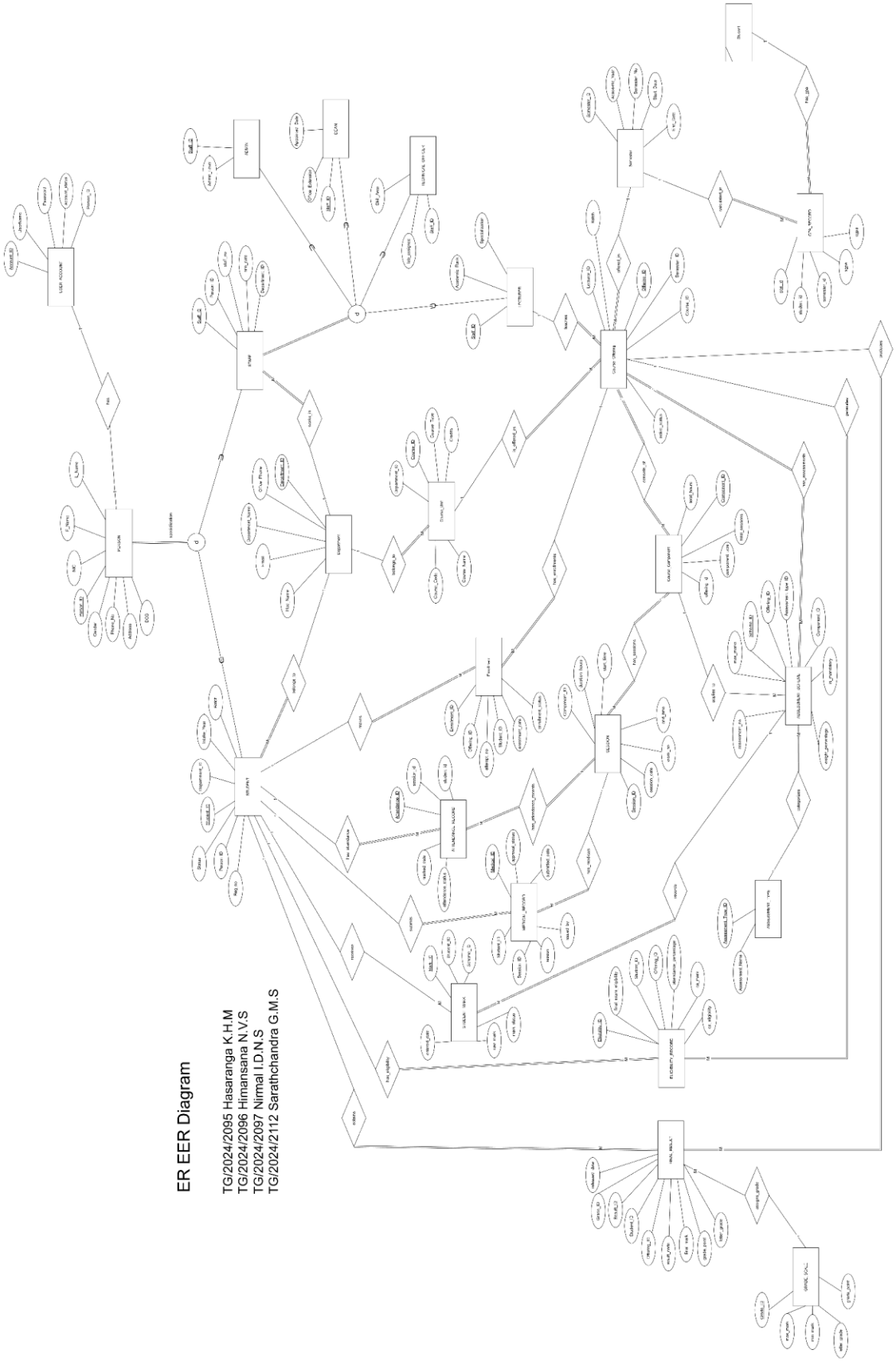
DEPARTMENT (Department_ID, Department_Name, Hod_Name, Email, Office_Phone) is a root entity. COURSE_UNIT is associated with a DEPARTMENT. A COURSE_OFFERING denotes a particular offering of a COURSE_UNIT during a particular SEMESTER, by a LECTURER, for a particular Batch. Each COURSE_OFFERING consists of one or more COURSE_COMPONENTs of type Theory/Practical. The ENROLLMENT entity handles the M:N relationship between STUDENT and COURSE_OFFERING along with enrollment_date, attempt_no, and enrollment_status.

3.3 Attendance Management

Each COURSE_COMPONENT has multiple SESSIONs (Session_ID, session_date, week_no, start_time, end_time, duration_hours). Per session, each enrolled STUDENT has an ATTENDANCE_RECORD (attendance_status: Present / Absent / Medical, marked_by linking to STAFF). Students may submit MEDICAL_RECORDs linked to specific absent sessions, with an approval_status field.

3.4 Marks & Results

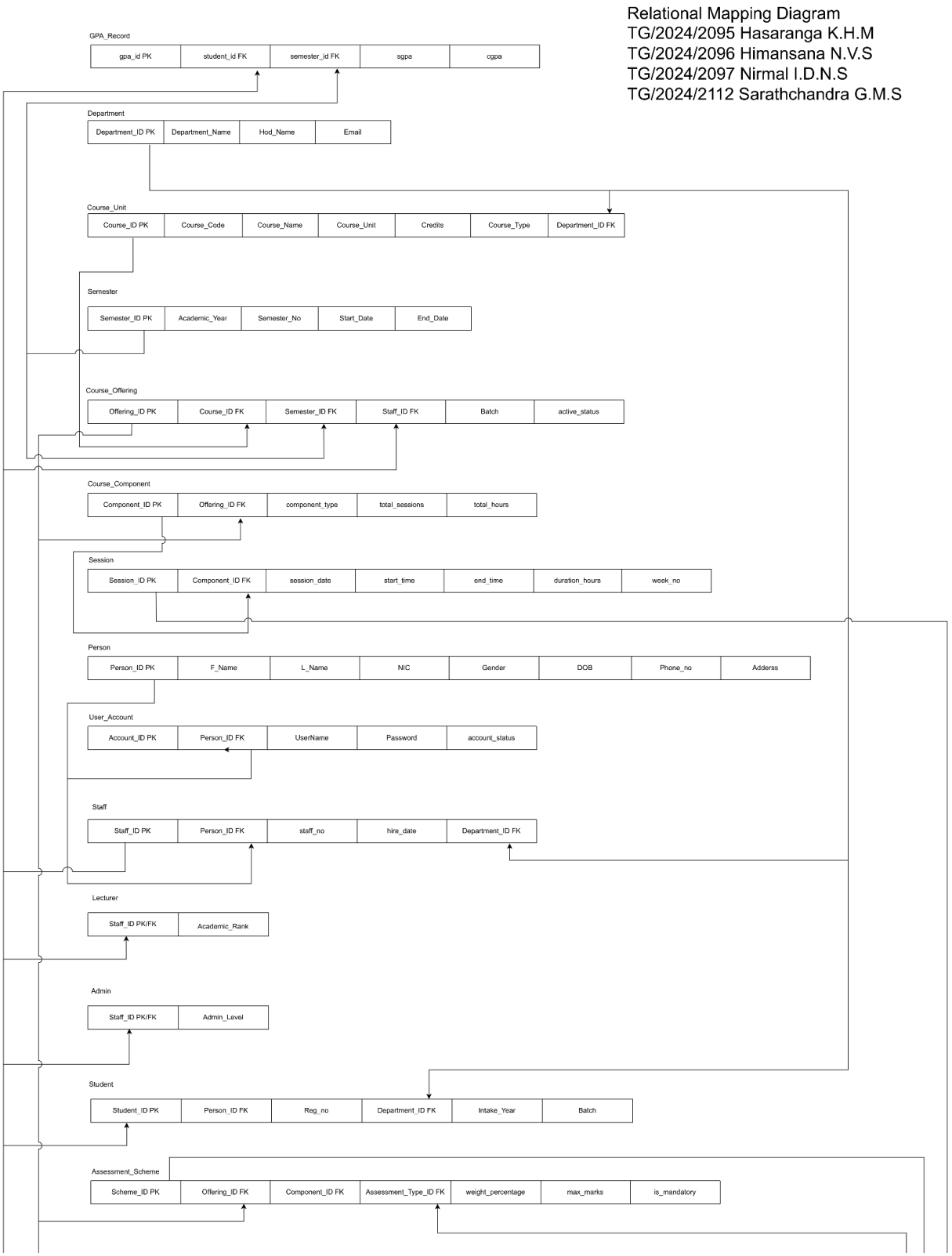
An ASSESSMENT_SCHEME defines the marking structure for each COURSE_OFFERING, linking to ASSESSMENT_TYPE (Quiz, Project, Mid Theory, Mid Practical, Final Theory, Final Practical) and optionally to a COURSE_COMPONENT. STUDENT_MARK stores the raw_mark entered by a LECTURER per student per scheme. ELIGIBILITY_RECORD is computed: attendance_percentage and ca_mark are derived, and eligibility flags are set. FINAL_RESULT stores the weighted final_mark and the letter_grade/grade_point assigned via GRADE_SCALE. GPA_RECORD holds the credit-weighted SGPA and CGPA per student per semester.

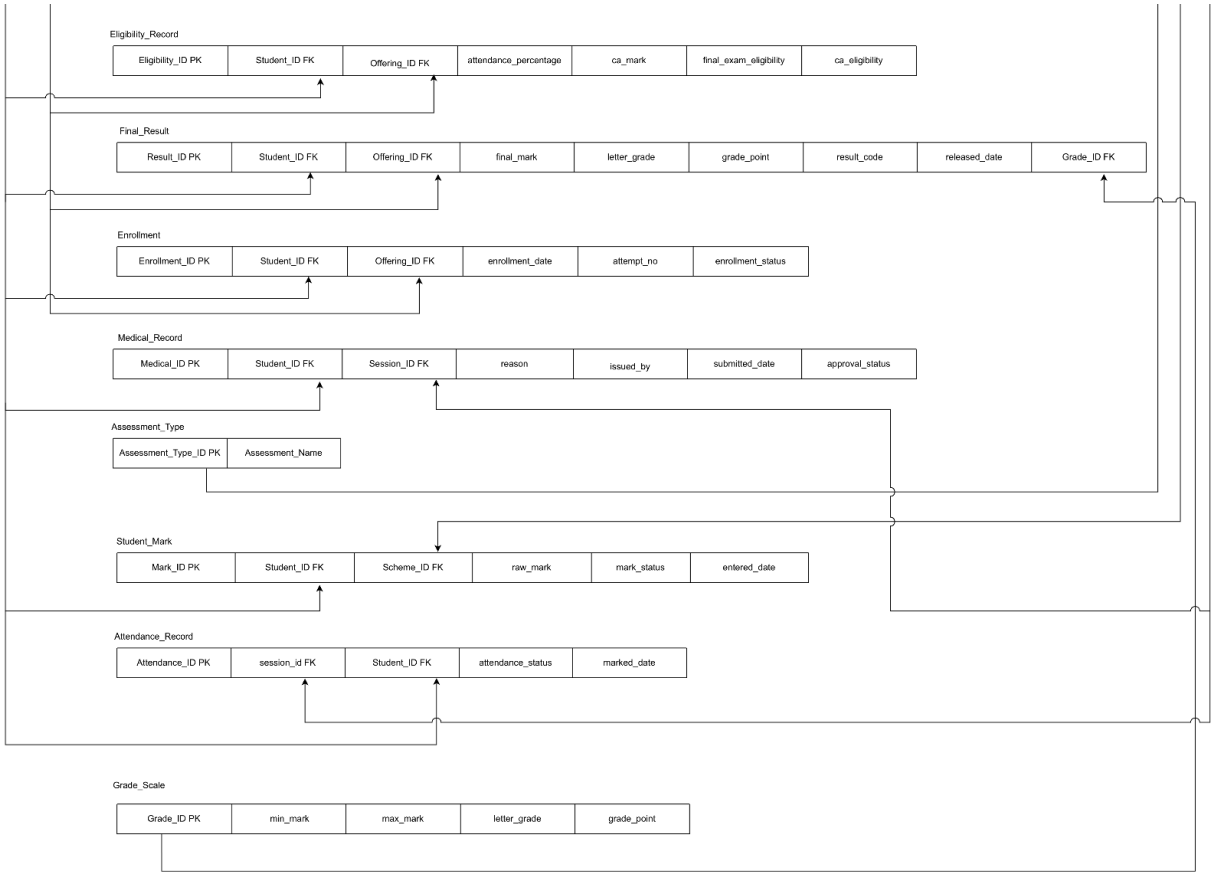


(Refer to the attached ER/EER diagram PDF for the complete visual diagram.) [link_ER_EER]

4. Proposed Relational Mapping Diagram

The relational mapping translates each entity and relationship from the ER/EER diagram into normalised relational tables (up to 3NF). The table below lists all 24 tables with their primary keys and key foreign key references.





(Refer to the attached relational mapping PDF for the complete visual diagram.) *link_relational_Mapping*

5. Table Structure of the Solution

This section presents the full DDL (CREATE TABLE) script ordered by dependency, followed by representative sample data inserts, and all database views with their actual SQL code.

5.1 Full Table Creation Script (Dependency-Ordered)

The following script creates all 24 tables in the correct order so that all foreign key references are satisfied. Each table is annotated with the team member who created it.

```
-- Created by: Nirmal (TG2097)
CREATE TABLE PERSON (
   Person_ID    VARCHAR(20) PRIMARY KEY,
   F_Name       VARCHAR(50)  NOT NULL,
   L_Name       VARCHAR(50)  NOT NULL,
   NIC          VARCHAR(20)  NOT NULL UNIQUE,
   Gender       ENUM('Male', 'Female', 'Other') NOT NULL,
   DOB          DATE         NOT NULL,
   Phone_No     VARCHAR(15),
   Address      VARCHAR(255)
);
```

```
+-----------+---------------+-------------+------------+--------+------------+------------+----------+
| Person_ID | F_Name        | L_Name      | NIC        | Gender | DOB        | Phone_No   | Address  |
+-----------+---------------+-------------+------------+--------+------------+------------+----------+
| P01       | Nalin         | Bandara     | 198011111V | Male   | 1980-01-10 | 0771000001 | Matara   |
| P02       | Saman         | Kumara      | 197522222V | Male   | 1975-05-20 | 0772000001 | Galle    |
| P03       | Amali         | Perera      | 198533331V | Female | 1985-02-15 | 0773000001 | Colombo  |
| P04       | Kasun         | Kalhara     | 198633332V | Male   | 1986-03-15 | 0773000002 | Matara   |
| P05       | Ruwan         | Pathirana   | 198733333V | Male   | 1987-04-15 | 0773000003 | Galle    |
| P06       | Nuwan         | Pradeep     | 198833334V | Male   | 1988-05-15 | 0773000004 | Tangalle |
| P07       | Sanduni       | Silva       | 198933335V | Female | 1989-06-15 | 0773000005 | Colombo  |
| P08       | Gayan         | Madushanka  | 199033336V | Male   | 1990-07-15 | 0773000006 | Matara   |
| P09       | Chathurika    | Peiris      | 199133337V | Female | 1991-08-15 | 0773000007 | Galle    |
| P10       | Nilantha      | Jayasooriya | 199233338V | Male   | 1992-09-15 | 0773000008 | Kandy    |
+-----------+---------------+-------------+------------+--------+------------+------------+----------+
40 rows in set (0.30 sec)
```

```
-- Created by: Samindi (TG2112)
CREATE TABLE DEPARTMENT(
   Department_ID VARCHAR(20) PRIMARY KEY NOT NULL ,
   Department  Name VARCHAR(100) NOT NULL,
   Office  Phone INT(11),
   Email VARCHAR(100),
   Hod_Name VARCHAR(100)
);
```

```
+---------------+----------------------------------------+--------------+----------------------+--------------------+
| Department_ID | Department_Name                        | Office_Phone | Email                | Hod_Name           |
+---------------+----------------------------------------+--------------+----------------------+--------------------+
| D01           | Information and Communication Technology |    412223334 | hod_ict@fot.ruh.ac.lk | Dr. Dinithi        |
| D02           | Engineering Technology                 |    412223335 | hod_et@fot.ruh.ac.lk  | Dr. Kamal Silva    |
| D03           | Biosystems Technology                  |    412223336 | hod_bst@fot.ruh.ac.lk | Dr. Nimal Fernando |
| D04           | Multidisciplinary Studies              |    412223337 | hod_mds@fot.ruh.ac.lk | Prof. Subash       |
+---------------+----------------------------------------+--------------+----------------------+--------------------+
4 rows in set (0.02 sec)
```

```
-- Created by: Samindi (TG2112)
CREATE TABLE SEMESTER(
   Semester_ID VARCHAR(20) PRIMARY KEY NOT NULL ,
   Academic  Year YEAR,
   Semester  No INT(5),
   Start  Date DATE,
   End_Date DATE
);
```

```
+-------------+---------------+-------------+------------+------------+
| Semester_ID | Academic_Year | Semester_No | Start_Date | End_Date   |
+-------------+---------------+-------------+------------+------------+
| SEM01       |          2024 |           1 | 2026-05-04 | 2026-08-20 |
+-------------+---------------+-------------+------------+------------+
1 row in set (0.01 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE ASSESSMENT  TYPE (
   Assessment  Type  ID VARCHAR(20) PRIMARY KEY,
   Assessment_Name VARCHAR(50)
);
```

```
+--------------------+-----------------+
| Assessment_Type_ID | Assessment_Name |
+--------------------+-----------------+
| AT_FP              | Final Practical |
| AT_FT              | Final Theory    |
| AT_MP              | Mid Practical   |
| AT_MT              | Mid Theory      |
| AT_PR              | Project         |
| AT_QZ              | Quiz            |
+--------------------+-----------------+
6 rows in set (0.02 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE GRADE_SCALE (
   Grade  ID VARCHAR(20) PRIMARY KEY,
   min  mark INT,
   max_mark INT,
   grade_point DECIMAL(3,2),
   letter  grade VARCHAR(5)
);
```

```
+----------+----------+----------+-------------+--------------+
| Grade_ID | min_mark | max_mark | grade_point | letter_grade |
+----------+----------+----------+-------------+--------------+
| G_A      |       75 |       84 |        4.00 | A            |
| G_AM     |       70 |       74 |        3.70 | A-           |
| G_AP     |       85 |      100 |        4.00 | A+           |
| G_B      |       60 |       64 |        3.00 | B            |
| G_BM     |       55 |       59 |        2.70 | B-           |
| G_BP     |       65 |       69 |        3.30 | B+           |
| G_C      |       45 |       49 |        2.00 | C            |
| G_CM     |       40 |       44 |        1.70 | C-           |
| G_CP     |       50 |       54 |        2.30 | C+           |
| G_D      |       35 |       39 |        1.30 | D            |
| G_E      |        0 |       34 |        0.00 | E            |
+----------+----------+----------+-------------+--------------+
11 rows in set (0.01 sec)
```

```
-- Created by: Nirmal (TG2097)
CREATE TABLE STAFF (
   Staff_ID       VARCHAR(20) NOT NULL,
   Person_ID      VARCHAR(20) NOT NULL,
   staff_no       VARCHAR(20) NOT NULL UNIQUE,
   hire_date      DATE        NOT NULL,
   Department_ID  VARCHAR(20) NOT NULL,
   PRIMARY KEY (Staff_ID),
   FOREIGN KEY (Person_ID)     REFERENCES PERSON(Person_ID),
   FOREIGN KEY (Department_ID) REFERENCES DEPARTMENT(Department_ID)
);
```

```
+----------+-----------+----------+------------+---------------+
| Staff_ID | Person_ID | staff_no | hire_date  | Department_ID |
+----------+-----------+----------+------------+---------------+
| ST01     | P01       | EMP001   | 2010-01-01 | D01           |
| ST02     | P02       | EMP002   | 2005-01-01 | D01           |
| ST03     | P03       | EMP003   | 2015-01-01 | D01           |
| ST04     | P04       | EMP004   | 2016-01-01 | D01           |
| ST05     | P05       | EMP005   | 2017-01-01 | D02           |
| ST06     | P06       | EMP006   | 2018-01-01 | D03           |
| ST07     | P07       | EMP007   | 2019-01-01 | D01           |
| ST08     | P08       | EMP008   | 2019-06-01 | D02           |
| ST09     | P09       | EMP009   | 2020-01-01 | D03           |
| ST10     | P10       | EMP010   | 2020-06-01 | D01           |
| ST11     | P11       | EMP011   | 2020-01-01 | D01           |
| ST12     | P12       | EMP012   | 2020-02-01 | D01           |
| ST13     | P13       | EMP013   | 2021-01-01 | D02           |
| ST14     | P14       | EMP014   | 2021-05-01 | D03           |
| ST15     | P15       | EMP015   | 2022-01-01 | D01           |
+----------+-----------+----------+------------+---------------+
15 rows in set (0.003 sec)
```

```
-- Created by: Nirmal (TG2097)
CREATE TABLE STUDENT (
   Student_ID    VARCHAR(20) NOT NULL,
   Person_ID     VARCHAR(20) NOT NULL,
   Reg_no        VARCHAR(20) NOT NULL UNIQUE,
   Department_ID VARCHAR(20) NOT NULL,
   Intake_Year   INT         NOT NULL,
   Batch         VARCHAR(20) NOT NULL,
   Status        VARCHAR(20),
   PRIMARY KEY (Student_ID),
   FOREIGN KEY (Person_ID)     REFERENCES PERSON(Person_ID),
   FOREIGN KEY (Department_ID) REFERENCES DEPARTMENT(Department_ID)
);
```

```
+------------+-----------+--------------+---------------+-------------+-------+--------+
| Student_ID | Person_ID | Reg_no       | Department_ID | Intake_Year | Batch | Status |
+------------+-----------+--------------+---------------+-------------+-------+--------+
| S01        | P16       | TG/2024/2095 | D01           |        2024 | 23/24 | Proper |
| S02        | P17       | TG/2024/2096 | D01           |        2024 | 23/24 | Proper |
| S03        | P18       | TG/2024/2097 | D01           |        2024 | 23/24 | Proper |
| S04        | P19       | TG/2024/2112 | D01           |        2024 | 23/24 | Proper |
| S05        | P20       | TG/2024/2113 | D01           |        2024 | 23/24 | Proper |
| S06        | P21       | TG/2024/2114 | D01           |        2024 | 23/24 | Proper |
| S07        | P22       | TG/2024/2115 | D01           |        2024 | 23/24 | Proper |
| S08        | P23       | TG/2024/2116 | D01           |        2024 | 23/24 | Proper |
| S09        | P24       | TG/2024/2117 | D01           |        2024 | 23/24 | Proper |
+------------+-----------+--------------+---------------+-------------+-------+--------+
25 rows in set (0.001 sec)
```

```
-- Created by: Nirmal (TG2097)
CREATE TABLE USER_ACCOUNT (
   Account_ID     VARCHAR(20)  NOT NULL,
   Person_ID      VARCHAR(20)  NOT NULL,
   UserName       VARCHAR(50)  NOT NULL UNIQUE,
   Password       VARCHAR(255) NOT NULL,
   account_status VARCHAR(20)  NOT NULL,
   PRIMARY KEY (Account_ID),
   FOREIGN KEY (Person_ID) REFERENCES PERSON(Person_ID)
);
```

```
MariaDB [fot_academic_management_db]> SELECT * FROM USER_ACCOUNT;+------------+-----------+--------------+----------+----------------
---+
| Account_ID | Person_ID | UserName     | Password | account_status |
+------------+-----------+--------------+----------+----------------+
| U01        | P01       | admin_nalin  | pass123  | Active         |
| U02        | P02       | dean_saman   | pass123  | Active         |
| U03        | P03       | lec_amali    | pass123  | Active         |
| U04        | P04       | lec_kasun    | pass123  | Active         |
| U05        | P05       | lec_ruwan    | pass123  | Active         |
| U06        | P06       | lec_nuwan    | pass123  | Active         |
| U07        | P07       | lec_sanduni  | pass123  | Active         |
| U08        | P08       | lec_gayan    | pass123  | Active         |
| U09        | P09       | lec_chathu   | pass123  | Active         |
| U10        | P10       | lec_nilantha | pass123  | Active         |
+------------+-----------+--------------+----------+----------------+
40 rows in set (0.002 sec)
```

```
-- Created by: Samindi  (TG2112)
CREATE TABLE COURSE_UNIT(
   Course  ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Course  Code VARCHAR(10) NOT NULL,
   Course_Name VARCHAR(50) NOT NULL,
   Credits INT,
   Course  Type VARCHAR(20),
   Department  ID VARCHAR(20),
   FOREIGN KEY (Department_ID) REFERENCES DEPARTMENT(Department_ID)
);
```

```
+------------+-------------+----------------------------------+---------+-------------+---------------+
| Course_ID  | Course_Code | Course_Name                      | Credits | Course_Type | Department_ID |
+------------+-------------+----------------------------------+---------+-------------+---------------+
| CU_ENG1212 | ENG1212     | English II                       |       2 | Theory      | D01           |
| CU_ICT1212 | ICT1212     | Database Management Systems      |       2 | Theory      | D01           |
| CU_ICT1222 | ICT1222     | DBMS Practicum                   |       2 | Practical   | D01           |
| CU_ICT1232 | ICT1232     | Web Development                  |       2 | Theory      | D01           |
| CU_ICT1242 | ICT1242     | Web Development Practicum        |       2 | Practical   | D01           |
| CU_ICT1252 | ICT1252     | OS Concepts and Application      |       2 | Theory      | D01           |
| CU_ICT1261 | ICT1261     | System Prog. Fundamentals & Linux |       2 | Both        | D01           |
| CU_TCS1212 | TCS1212     | Fundamentals of Management       |       2 | Theory      | D01           |
| CU_TMS1233 | TMS1233     | Discrete Mathematics             |       3 | Theory      | D01           |
+------------+-------------+----------------------------------+---------+-------------+---------------+
9 rows in set (0.001 sec)
```

```sql
-- Created by: Nirmal (TG2097)
CREATE TABLE ADMIN (
   Staff_ID    VARCHAR(20) NOT NULL,
   Admin  Level VARCHAR(50) NOT NULL,
   PRIMARY KEY (Staff  ID),
   FOREIGN KEY (Staff  ID) REFERENCES STAFF(Staff  ID)
);
```

```
+----------+-------------+
| Staff_ID | Admin_Level |
+----------+-------------+
| ST01     | Super Admin |
+----------+-------------+
1 row in set (0.004 sec)
```

```sql
-- Created by: Nirmal (TG2097)
CREATE TABLE DEAN (
   Staff_ID       VARCHAR(20) NOT NULL,
   Appointed_Date  DATE       NOT NULL,
   PRIMARY KEY (Staff_ID),
   FOREIGN KEY (Staff_ID) REFERENCES STAFF(Staff_ID)
);
```

```
+----------+----------------+
| Staff_ID | Appointed_Date |
+----------+----------------+
| ST02     | 2024-01-01     |
+----------+----------------+
1 row in set (0.003 sec)
```

```sql
-- Created by: Nirmal (TG2097)
CREATE TABLE LECTURER (
   Staff  ID      VARCHAR(20) NOT NULL,
   Academic_Rank VARCHAR(50) NOT NULL,
   PRIMARY KEY (Staff_ID),
   FOREIGN KEY (Staff_ID) REFERENCES STAFF(Staff_ID)
);
```

```
+----------+-----------------+
| Staff_ID | Academic_Rank   |
+----------+-----------------+
| ST03     | Senior Lecturer |
| ST04     | Lecturer        |
| ST05     | Senior Lecturer |
| ST06     | Lecturer        |
| ST07     | Lecturer        |
| ST08     | Senior Lecturer |
| ST09     | Lecturer        |
| ST10     | Lecturer        |
+----------+-----------------+
8 rows in set (0.003 sec)
```

```sql
-- Created by: Nirmal (TG2097)
CREATE TABLE TECHNICAL_OFFICER (
    Staff_ID         VARCHAR(20) NOT NULL,
    Specialization   VARCHAR(100) NOT NULL,
    lab_assigned     VARCHAR(100),
    PRIMARY KEY (Staff_ID),
    FOREIGN KEY (Staff_ID) REFERENCES STAFF(Staff_ID)
);
```

```
+----------+----------------+--------------+
| Staff_ID | Specialization | lab_assigned |
+----------+----------------+--------------+
| ST11     | Networking     | Lab A        |
| ST12     | Hardware       | Lab B        |
| ST13     | Software       | Lab C        |
| ST14     | Hardware       | Lab D        |
| ST15     | Networking     | Lab E        |
+----------+----------------+--------------+
5 rows in set (0.004 sec)
```

```sql
-- Created by: Hasaranga (TG2095)
CREATE TABLE COURSE_OFFERING (
   Offering  ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Course  ID VARCHAR(20) NOT NULL,
   Lecturer  ID VARCHAR(20) NOT NULL,
   Semester_ID VARCHAR(20) NOT NULL,
   Batch VARCHAR(10),
   active  status BOOLEAN,
   FOREIGN KEY (Course  ID) REFERENCES COURSE  UNIT(Course  ID),
   FOREIGN KEY (Lecturer_ID) REFERENCES LECTURER(Staff_ID),
   FOREIGN KEY (Semester  ID) REFERENCES SEMESTER(Semester  ID)
);
```

```
+-------------+------------+-------------+-------------+-----------+---------------+
| Offering_ID | Course_ID  | Lecturer_ID | Semester_ID | Batch     | active_status |
+-------------+------------+-------------+-------------+-----------+---------------+
| OFF_DBP     | CU_ICT1222 | ST05        | SEM01       | 2022/2023 |             1 |
| OFF_DBS     | CU_ICT1212 | ST10        | SEM01       | 2022/2023 |             1 |
| OFF_ENG     | CU_ENG1212 | ST09        | SEM01       | 2022/2023 |             1 |
| OFF_LIN     | CU_ICT1261 | ST04        | SEM01       | 2022/2023 |             1 |
| OFF_MAT     | CU_TMS1233 | ST08        | SEM01       | 2022/2023 |             1 |
| OFF_MGT     | CU_TCS1212 | ST03        | SEM01       | 2022/2023 |             1 |
| OFF_OSC     | CU_ICT1252 | ST09        | SEM01       | 2022/2023 |             1 |
| OFF_WEB     | CU_ICT1232 | ST06        | SEM01       | 2022/2023 |             1 |
| OFF_WEP     | CU_ICT1242 | ST07        | SEM01       | 2022/2023 |             1 |
+-------------+------------+-------------+-------------+-----------+---------------+
9 rows in set (0.000 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE COURSE  COMPONENT (
   Component_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   offering  id VARCHAR(20) NOT NULL,
   component  type VARCHAR(20),
   total  sessions INT,
   total_hours INT,
   FOREIGN KEY (offering  id) REFERENCES COURSE  OFFERING(Offering  ID)
);
```

```
+--------------+------------+---------------+----------------+-------------+
| Component_ID | offering_id | component_type | total_sessions | total_hours |
+--------------+------------+---------------+----------------+-------------+
| CC_DBP       | OFF_DBP     | Practical      |             15 |          30 |
| CC_DBS       | OFF_DBS     | Theory         |             15 |          30 |
| CC_ENG       | OFF_ENG     | Theory         |             15 |          30 |
| CC_LIN_P     | OFF_LIN     | Practical      |             15 |          15 |
| CC_LIN_T     | OFF_LIN     | Theory         |             15 |          15 |
| CC_MAT       | OFF_MAT     | Theory         |             15 |          30 |
| CC_MGT       | OFF_MGT     | Theory         |             15 |          30 |
| CC_OSC       | OFF_OSC     | Theory         |             15 |          30 |
| CC_WEB_P1    | OFF_WEP     | Practical      |             15 |          30 |
| CC_WEB_P2    | OFF_WEP     | Practical      |             15 |          45 |
| CC_WEB_T     | OFF_WEB     | Theory         |             15 |          15 |
+--------------+------------+---------------+----------------+-------------+
11 rows in set (0.001 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE ENROLLMENT (
   Enrollment_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Student_ID VARCHAR(20) NOT NULL,
   Offering  ID VARCHAR(20) NOT NULL,
   enrollment  date DATE,
   attempt_no INT,
   enrollment_status VARCHAR(20),
   FOREIGN KEY (Student  ID) REFERENCES STUDENT(Student  ID),
   FOREIGN KEY (Offering  ID) REFERENCES COURSE  OFFERING(Offering  ID)
);
```

```
+---------------+------------+-------------+-----------------+------------+-------------------+
| Enrollment_ID | Student_ID | Offering_ID | enrollment_date | attempt_no | enrollment_status |
+---------------+------------+-------------+-----------------+------------+-------------------+
| E_DBP_01      | S01        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_02      | S02        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_03      | S03        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_04      | S04        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_05      | S05        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_06      | S06        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_07      | S07        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_08      | S08        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_09      | S09        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_10      | S10        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_11      | S11        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_12      | S12        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_13      | S13        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_14      | S14        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_15      | S15        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_16      | S16        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_17      | S17        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_18      | S18        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_19      | S19        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
| E_DBP_20      | S20        | OFF_DBP     | 2026-05-01      |          1 | Enrolled          |
+---------------+------------+-------------+-----------------+------------+-------------------+
225 rows in set (0.004 sec)
```

```
-- Created by: Samindi (TG2112)
CREATE TABLE SESSION (
   Session_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Component_ID VARCHAR(20) NOT NULL,
   session_date DATE,
   week_no INT,
   start_time TIME,
   end_time TIME,
   duration_hours INT,
   FOREIGN KEY (Component_ID) REFERENCES COURSE_COMPONENT(Component_ID)
);
```

```
+------------+--------------+--------------+---------+------------+----------+----------------+
| Session_ID | Component_ID | session_date | week_no | start_time | end_time | duration_hours |
+------------+--------------+--------------+---------+------------+----------+----------------+
| W01_DBP    | CC_DBP       | 2026-05-05   |       1 | 09:00:00   | 12:00:00 |              3 |
| W01_DBS    | CC_DBS       | 2026-05-07   |       1 | 10:00:00   | 12:00:00 |              2 |
| W01_ENG    | CC_ENG       | 2026-05-07   |       1 | 09:00:00   | 10:00:00 |              1 |
| W01_LIN    | CC_LIN_T     | 2026-05-04   |       1 | 13:00:00   | 15:00:00 |              2 |
| W01_MAT    | CC_MAT       | 2026-05-06   |       1 | 09:00:00   | 12:00:00 |              3 |
| W01_MGT    | CC_MGT       | 2026-05-04   |       1 | 09:00:00   | 11:00:00 |              2 |
| W01_OSC    | CC_OSC       | 2026-05-08   |       1 | 08:00:00   | 10:00:00 |              2 |
| W01_WEBP1  | CC_WEB_P1    | 2026-05-05   |       1 | 14:00:00   | 16:00:00 |              2 |
| W01_WEBP2  | CC_WEB_P2    | 2026-05-07   |       1 | 13:00:00   | 16:00:00 |              3 |
| W01_WEBT   | CC_WEB_T     | 2026-05-05   |       1 | 13:00:00   | 14:00:00 |              1 |
| W02_DBP    | CC_DBP       | 2026-05-12   |       2 | 09:00:00   | 12:00:00 |              3 |
| W02_DBS    | CC_DBS       | 2026-05-14   |       2 | 10:00:00   | 12:00:00 |              2 |
| W02_ENG    | CC_ENG       | 2026-05-14   |       2 | 09:00:00   | 10:00:00 |              1 |
| W02_LIN    | CC_LIN_T     | 2026-05-11   |       2 | 13:00:00   | 15:00:00 |              2 |
| W02_MAT    | CC_MAT       | 2026-05-13   |       2 | 09:00:00   | 12:00:00 |              3 |
| W02_MGT    | CC_MGT       | 2026-05-11   |       2 | 09:00:00   | 11:00:00 |              2 |
+------------+--------------+--------------+---------+------------+----------+----------------+
150 rows in set (0.004 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE GPA_RECORD (
   gpa_id VARCHAR(20) PRIMARY KEY NOT NULL,
   student_id VARCHAR(20) NOT NULL,
   semester_id VARCHAR(20) NOT NULL,
   sgpa DECIMAL(4,2),
   cgpa DECIMAL(4,2),
   FOREIGN KEY (student_id) REFERENCES STUDENT(Student_ID),
   FOREIGN KEY (semester_id) REFERENCES SEMESTER(Semester_ID)
);
```

```
+---------------+------------+-------------+------+------+
| gpa_id        | student_id | semester_id | sgpa | cgpa |
+---------------+------------+-------------+------+------+
| GPA_SEM01_S01 | S01        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S02 | S02        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S03 | S03        | SEM01       | 3.76 | 3.76 |
| GPA_SEM01_S04 | S04        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S05 | S05        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S06 | S06        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S07 | S07        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S08 | S08        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S09 | S09        | SEM01       | 0.00 | 0.00 |
| GPA_SEM01_S10 | S10        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S11 | S11        | SEM01       | 3.91 | 3.91 |
| GPA_SEM01_S12 | S12        | SEM01       | 4.00 | 4.00 |
| GPA_SEM01_S13 | S13        | SEM01       | 0.00 | 0.00 |
+---------------+------------+-------------+------+------+
25 rows in set (0.001 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE ASSESSMENT  SCHEME (
   Scheme_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Offering  ID VARCHAR(20) NOT NULL,
   Assessment  Type  ID VARCHAR(20) NOT NULL,
   Component  ID VARCHAR(20),
   weight_percentage DECIMAL(5,2),
   assessment  no INT,
   max  marks INT,
   is  mandatory BOOLEAN,
   FOREIGN KEY (Offering_ID) REFERENCES COURSE_OFFERING(Offering_ID),
   FOREIGN KEY (Assessment  Type  ID) REFERENCES ASSESSMENT  TYPE(Assessment  Type  ID),
   FOREIGN KEY (Component  ID) REFERENCES COURSE  COMPONENT(Component  ID)
);
```

Scheme_ID	Offering_ID	Assessment_Type_ID	Component_ID	weight_percentage	assessment_no	max_marks	is_mandatory
SC_DBP_FP	OFF_DBP	AT_FP	CC_DBP	60.00	1	100	1
SC_DBP_MP	OFF_DBP	AT_MP	CC_DBP	20.00	1	100	1
SC_DBP_MT	OFF_DBP	AT_MT	CC_DBP	20.00	1	100	1
SC_DBP_PR	OFF_DBP	AT_PR	CC_DBP	10.00	1	100	1
SC_DBP_QZ	OFF_DBP	AT_QZ	CC_DBP	10.00	1	100	1
SC_DBS_FT	OFF_DBS	AT_FT	CC_DBS	70.00	1	100	1
SC_DBS_MT	OFF_DBS	AT_MT	CC_DBS	15.00	1	100	1
SC_DBS_PR	OFF_DBS	AT_PR	CC_DBS	10.00	1	100	1
SC_DBS_QZ	OFF_DBS	AT_QZ	CC_DBS	5.00	1	100	1
SC_ENG_FT	OFF_ENG	AT_FT	CC_ENG	70.00	1	100	1
SC_ENG_MT	OFF_ENG	AT_MT	CC_ENG	15.00	1	100	1
SC_ENG_PR	OFF_ENG	AT_PR	CC_ENG	10.00	1	100	1

38 rows in set (0.001 sec)

```
-- Created by: Samindi  (TG2112)
CREATE TABLE ATTENDANCE  RECORD (
   Attendance  ID VARCHAR(20) PRIMARY KEY NOT NULL,
   session  id VARCHAR(20) NOT NULL,
   student_id VARCHAR(20) NOT NULL,
   marked  by VARCHAR(20) NOT NULL,
   attendance  status VARCHAR(20),
   marked  date DATE,
   FOREIGN KEY (session_id) REFERENCES SESSION(Session_ID),
   FOREIGN KEY (student  id) REFERENCES STUDENT(Student  ID),
   FOREIGN KEY (marked  by) REFERENCES STAFF(Staff  ID)
);
```

Attendance_ID	session_id	student_id	marked_by	attendance_status	marked_date
A_W01_DBP_01	W01_DBP	S01	ST05	Present	2026-05-05
A_W01_DBP_02	W01_DBP	S02	ST05	Present	2026-05-05
A_W01_DBP_03	W01_DBP	S03	ST05	Present	2026-05-05
A_W01_DBP_04	W01_DBP	S04	ST05	Present	2026-05-05
A_W01_DBP_05	W01_DBP	S05	ST05	Present	2026-05-05
A_W01_DBP_06	W01_DBP	S06	ST05	Present	2026-05-05
A_W01_DBP_07	W01_DBP	S07	ST05	Present	2026-05-05
A_W01_DBP_08	W01_DBP	S08	ST05	Present	2026-05-05
A_W01_DBP_09	W01_DBP	S09	ST05	Present	2026-05-05
A_W01_DBP_10	W01_DBP	S10	ST05	Present	2026-05-05
A_W01_DBS_01	W01_DBS	S01	ST10	Present	2026-05-07

```
| A_W01_DBS_02   | W01_DBS    | S02        | ST10      | Present           | 2026-05-07  |
| A_W01_DBS_03   | W01_DBS    | S03        | ST10      | Present           | 2026-05-07  |
| A_W01_DBS_04   | W01_DBS    | S04        | ST10      | Medical           | 2026-05-07  |
| A_W01_DBS_05   | W01_DBS    | S05        | ST10      | Present           | 2026-05-07  |
| A_W01_DBS_06   | W01_DBS    | S06        | ST10      | Present           | 2026-05-07  |
| A_W01_DBS_07   | W01_DBS    | S07        | ST10      | Present           | 2026-05-07  |
+----------------+------------+------------+-----------+-------------------+-------------+
3000 rows in set (0.021 sec)
```

```sql
-- Created by: Samindi  (TG2112)
CREATE TABLE MEDICAL_RECORD (
   Medical_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Student_ID VARCHAR(20) NOT NULL,
   Session_ID VARCHAR(20) NOT NULL,
   reason VARCHAR(255),
   issued_by VARCHAR(100),
   submitted_date DATE,
   approval_status VARCHAR(20),
   FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID),
   FOREIGN KEY (Session_ID) REFERENCES SESSION(Session_ID)
);
```

```
+------------+------------+------------+----------------------+---------------------------+---------------+-----------------+
| Medical_ID | Student_ID | Session_ID | reason               | issued_by                 | submitted_date | approval_status |
+------------+------------+------------+----------------------+---------------------------+---------------+-----------------+
| MED_001    | S04        | W01_DBS    | Viral Fever          | General Hospital Matara   | 2026-05-10     | Approved        |
| MED_002    | S04        | W02_LIN    | Dengue Fever         | General Hospital Matara   | 2026-05-15     | Approved        |
| MED_003    | S07        | W03_MAT    | Severe Migraine      | University Medical Center | 2026-05-25     | Approved        |
| MED_004    | S09        | W06_LIN    | Food Poisoning       | Dr. Perera (Private Clinic) | 2026-06-10     | Approved        |
| MED_005    | S08        | W10_MGT    | Sports Injury (Ankle) | Sports Medical Unit       | 2026-07-08     | Pending         |
+------------+------------+------------+----------------------+---------------------------+---------------+-----------------+
5 rows in set (0.001 sec)
```

```sql
-- Created by: Hasaranga (TG2095)
CREATE TABLE STUDENT_MARK (
   Mark_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Student_ID VARCHAR(20) NOT NULL,
   Scheme_ID VARCHAR(20) NOT NULL,
   entered_by VARCHAR(20) NOT NULL,
   raw_mark DECIMAL(5,2),
   mark_status VARCHAR(20),
   entered_date DATE,
   FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID),
   FOREIGN KEY (Scheme_ID) REFERENCES ASSESSMENT_SCHEME(Scheme_ID),
   FOREIGN KEY (entered_by) REFERENCES STAFF(Staff_ID)
);
```

```
+-----------+------------+-----------+------------+----------+-------------+--------------+
| Mark_ID   | Student_ID | Scheme_ID | entered_by | raw_mark | mark_status | entered_date |
+-----------+------------+-----------+------------+----------+-------------+--------------+
| M_DBPF_01 | S01        | SC_DBP_FP | ST05       |    90.00 | Verified    | 2026-08-20   |
| M_DBPF_02 | S02        | SC_DBP_FP | ST05       |    86.00 | Verified    | 2026-08-20   |
| M_DBPF_03 | S03        | SC_DBP_FP | ST05       |    80.00 | Verified    | 2026-08-20   |
| M_DBPF_06 | S06        | SC_DBP_FP | ST05       |    82.00 | Verified    | 2026-08-20   |
| M_DBPF_07 | S07        | SC_DBP_FP | ST05       |    90.00 | Verified    | 2026-08-20   |
| M_DBPF_08 | S08        | SC_DBP_FP | ST05       |    94.00 | Verified    | 2026-08-20   |
| M_DBPF_09 | S09        | SC_DBP_FP | ST05       |     0.00 | Verified    | 2026-08-20   |
| M_DBPF_10 | S10        | SC_DBP_FP | ST05       |    85.00 | Verified    | 2026-08-20   |
+-----------+------------+-----------+------------+----------+-------------+--------------+
900 rows in set (0.002 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE ELIGIBILITY_RECORD (
   Eligibility_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Student_ID VARCHAR(20) NOT NULL,
   Offering_ID VARCHAR(20) NOT NULL,
   attendance_percentage DECIMAL(5,2),
   ca_mark DECIMAL(5,2),
   ca_eligibility BOOLEAN,
   final_exam_eligibility BOOLEAN,
   FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID),
   FOREIGN KEY (Offering_ID) REFERENCES COURSE_OFFERING(Offering_ID)
);
```

```
+-----------------+------------+-------------+----------------------+---------+---------------+------------------------+
| Eligibility_ID  | Student_ID | Offering_ID | attendance_percentage | ca_mark | ca_eligibility | final_exam_eligibility |
+-----------------+------------+-------------+----------------------+---------+---------------+------------------------+
| ELI_OFF_DBP_S01 | S01        | OFF_DBP     |                100.00 |   34.30 |             1 |                      1 |
| ELI_OFF_DBP_S02 | S02        | OFF_DBP     |                100.00 |   32.60 |             1 |                      1 |
| ELI_OFF_DBP_S03 | S03        | OFF_DBP     |                100.00 |   29.40 |             1 |                      1 |
| ELI_OFF_DBP_S04 | S04        | OFF_DBP     |                100.00 |   37.00 |             1 |                      1 |
| ELI_OFF_DBP_S05 | S05        | OFF_DBP     |                100.00 |   37.90 |             1 |                      1 |
| ELI_OFF_DBP_S06 | S06        | OFF_DBP     |                100.00 |   30.90 |             1 |                      1 |
| ELI_OFF_DBP_S07 | S07        | OFF_DBP     |                100.00 |   34.70 |             1 |                      1 |
| ELI_OFF_DBP_S08 | S08        | OFF_DBP     |                100.00 |   36.00 |             1 |                      1 |
| ELI_OFF_DBP_S09 | S09        | OFF_DBP     |                100.00 |    0.00 |             1 |                      1 |
| ELI_OFF_DBP_S10 | S10        | OFF_DBP     |                100.00 |   31.80 |             1 |                      1 |
+-----------------+------------+-------------+----------------------+---------+---------------+------------------------+
225 rows in set (0.001 sec)
```

```
-- Created by: Hasaranga (TG2095)
CREATE TABLE FINAL_RESULT (
   Result_ID VARCHAR(20) PRIMARY KEY NOT NULL,
   Student_ID VARCHAR(20) NOT NULL,
   Offering_ID VARCHAR(20) NOT NULL,
   final_mark DECIMAL(5,2),
   letter_grade VARCHAR(5),
   grade_point DECIMAL(3,2),
   result_code VARCHAR(10),
   released_date DATE,
   FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID),
   FOREIGN KEY (Offering_ID) REFERENCES COURSE_OFFERING(Offering_ID)
);
```

```
+-----------------+------------+-------------+------------+--------------+-------------+-------------+---------------+
| Result_ID       | Student_ID | Offering_ID | final_mark | letter_grade | grade_point | result_code | released_date |
+-----------------+------------+-------------+------------+--------------+-------------+-------------+---------------+
| RES_OFF_DBP_S01 | S01        | OFF_DBP     |      88.30 | A+           |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S02 | S02        | OFF_DBP     |      84.20 | A            |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S03 | S03        | OFF_DBP     |      77.40 | A            |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S04 | S04        | OFF_DBP     |      94.00 | A+           |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S05 | S05        | OFF_DBP     |      96.70 | A+           |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S06 | S06        | OFF_DBP     |      80.10 | A            |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S07 | S07        | OFF_DBP     |      88.70 | A+           |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S08 | S08        | OFF_DBP     |      92.40 | A+           |        4.00 | PASS        | 2026-05-07    |
| RES_OFF_DBP_S09 | S09        | OFF_DBP     |       0.00 | E            |        0.00 | FAIL        | 2026-05-07    |
| RES_OFF_DBP_S10 | S10        | OFF_DBP     |      82.80 | A            |        4.00 | PASS        | 2026-05-07    |
+-----------------+------------+-------------+------------+--------------+-------------+-------------+---------------+
225 rows in set (0.001 sec)
```

5.2 Database Views

The system defines 15 views across four categories. Students and Technical Officers access data exclusively through these views they have no direct access to base tables.

ATTENDANCE VIEWS created by Hasaranga (TG2095)

1. Batch_Attendance_Summary overall attendance % and eligibility per student per course.

```
CREATE VIEW Batch Attendance Summary AS
SELECT
   ST.Reg No,
   CU.Course_Code,
   CU.Course Name,
   COUNT(S.Session ID) AS Total Sessions,
   SUM(CASE WHEN A.attendance_status IN ('Present', 'Medical') THEN 1 ELSE 0 END) AS Attended_Sessions,
   ROUND((SUM(CASE WHEN A.attendance_status IN ('Present', 'Medical') THEN 1 ELSE 0 END) /
COUNT(S.Session ID)) * 100, 2) AS Attendance Percentage,
   CASE
      WHEN (SUM(CASE WHEN A.attendance_status IN ('Present', 'Medical') THEN 1 ELSE 0 END) /
COUNT(S.Session_ID)) * 100 >= 80.00 THEN 'Eligible'
      ELSE 'Not Eligible'
   END AS Eligibility Status
FROM STUDENT ST
JOIN ATTENDANCE_RECORD A ON ST.Student_ID = A.student_id
JOIN SESSION S ON A.session id = S.Session ID
JOIN COURSE_COMPONENT CC ON S.Component_ID = CC.Component_ID
JOIN COURSE_OFFERING CO ON CC.offering_id = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
GROUP BY ST.Reg_No, CU.Course_Code, CU.Course_Name;
```

2. Component_Wise_Attendance attendance split by Theory / Practical component.

```
CREATE VIEW Component_Wise_Attendance AS
SELECT
   ST.Reg No,
   CU.Course Code,
   CC.component_type,
   COUNT(S.Session_ID) AS Total_Sessions,
   SUM(CASE WHEN A.attendance status IN ('Present', 'Medical') THEN 1 ELSE 0 END) AS Attended Sessions,
   ROUND((SUM(CASE WHEN A.attendance_status IN ('Present', 'Medical') THEN 1 ELSE 0 END) /
COUNT(S.Session_ID)) * 100, 2) AS Component_Attendance_Percentage
FROM STUDENT ST
JOIN ATTENDANCE RECORD A ON ST.Student ID = A.student id
JOIN SESSION S ON A.session_id = S.Session_ID
JOIN COURSE_COMPONENT CC ON S.Component_ID = CC.Component_ID
JOIN COURSE_OFFERING CO ON CC.offering_id = CO.Offering_ID
JOIN COURSE UNIT CU ON CO.Course ID = CU.Course ID
GROUP BY ST.Reg_No, CU.Course_Code, CC.component_type;
```

3. Individual_Attendance_Summary attendance % per student per subject with name.

```
CREATE VIEW Individual_Attendance_Summary AS
SELECT
   ST.Reg_No,
   P.F_Name,
   P.L_Name,
   CU.Course_Code,
   ROUND((SUM(CASE WHEN A.attendance_status IN ('Present', 'Medical') THEN 1 ELSE 0 END) /
COUNT(S.Session_ID)) * 100, 2) AS Attendance_Percentage
FROM STUDENT ST
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN ATTENDANCE_RECORD A ON ST.Student_ID = A.student_id
JOIN SESSION S ON A.session_id = S.Session_ID
JOIN COURSE_COMPONENT CC ON S.Component_ID = CC.Component_ID
JOIN COURSE_OFFERING CO ON CC.offering_id = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
GROUP BY ST.Reg_No, P.F_Name, P.L_Name, CU.Course_Code;
```

4. Individual_Subject_Attendance_Detail session-by-session attendance record per student.

```
CREATE VIEW Individual Subject Attendance Detail AS
SELECT
   ST.Reg_No,
   CU.Course Code,
   S.session date,
   S.start time,
   CC.component_type,
   A.attendance status,
   A.marked date
FROM STUDENT ST
JOIN ATTENDANCE_RECORD A ON ST.Student_ID = A.student_id
JOIN SESSION S ON A.session id = S.Session ID
JOIN COURSE COMPONENT CC ON S.Component ID = CC.Component ID
JOIN COURSE OFFERING CO ON CC.offering id = CO.Offering ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID;
```

MARKS VIEWS created by Samindi (TG2112)

5. Individual_CA_Marks_Detail each CA mark with weight, excluding final exams (AT_FT, AT_FP).

```
CREATE VIEW Individual_CA_Marks_Detail AS
SELECT
   ST.Reg_No,
   CU.Course_Code,
   ATY.Assessment_Name,
   SM.raw_mark,
   SCH.weight_percentage,
   ROUND((SM.raw_mark * (SCH.weight_percentage / 100)), 2) AS weighted_mark
FROM STUDENT_MARK SM
JOIN ASSESSMENT_SCHEME SCH ON SM.Scheme_ID = SCH.Scheme_ID
JOIN ASSESSMENT_TYPE ATY ON SCH.Assessment_Type_ID = ATY.Assessment_Type_ID
JOIN STUDENT ST ON SM.Student_ID = ST.Student_ID
JOIN COURSE_OFFERING CO ON SCH.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
WHERE ATY.Assessment_Type_ID NOT IN ('AT_FT', 'AT_FP');
```

6. Student_CA_Marks_Summary total CA mark and CA eligibility per student per course.

```
CREATE VIEW Student_CA_Marks_Summary AS
SELECT
    ST.Reg_No,
    P.F_Name,
    P.L_Name,
    CU.Course_Code,
    CU.Course_Name,
    ER.ca_mark,
    ER.ca_eligibility
FROM STUDENT ST
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN ELIGIBILITY_RECORD ER ON ST.Student_ID = ER.Student_ID
JOIN COURSE_OFFERING CO ON ER.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID;
```

7. Batch_CA_Marks_Summary CA marks for the whole batch, ordered by course then registration number.

```
CREATE VIEW Batch_CA_Marks_Summary AS
SELECT
    CU.Course_Code,
    CU.Course_Name,
    ST.Reg_No,
    P.F_Name,
    P.L_Name,
    ER.ca_mark,
    ER.ca_eligibility
```

```
FROM ELIGIBILITY_RECORD ER
JOIN STUDENT ST ON ER.Student_ID = ST.Student_ID
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN COURSE_OFFERING CO ON ER.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
ORDER BY CU.Course_Code, ST.Reg_No;
```

8. Comprehensive_Marks_Breakdown all assessment marks pivoted to columns per student per course.

```
   CREATE VIEW Comprehensive_Marks_Breakdown AS
    SELECT
        ST.Reg_No,
        CU.Course_Code,
        CU.Course_Name,
        -- Quiz Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_QZ' THEN SM.raw_mark ELSE NULL END)
AS Quiz_Mark,
        -- Project Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_PR' THEN SM.raw_mark ELSE NULL END)
AS Project_Mark,
        -- Mid Theory Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_MT' THEN SM.raw_mark ELSE NULL END)
AS Mid_Theory_Mark,
        -- Mid Practical Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_MP' THEN SM.raw_mark ELSE NULL END)
AS Mid_Practical_Mark,
        -- Final Theory Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_FT' THEN SM.raw_mark ELSE NULL END)
AS Final_Theory_Mark,
        -- Final Practical Marks
        MAX(CASE WHEN ATY.Assessment_Type_ID = 'AT_FP' THEN SM.raw_mark ELSE NULL END)
AS Final_Practical_Mark
    FROM STUDENT ST
    JOIN STUDENT_MARK SM ON ST.Student_ID = SM.Student_ID
    JOIN ASSESSMENT_SCHEME SCH ON SM.Scheme_ID = SCH.Scheme_ID
    JOIN ASSESSMENT_TYPE ATY ON SCH.Assessment_Type_ID = ATY.Assessment_Type_ID
    JOIN COURSE_OFFERING CO ON SCH.Offering_ID = CO.Offering_ID
    JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
    GROUP BY ST.Reg_No, CU.Course_Code, CU.Course_Name;
```

9. Final_Marks_Summary attendance %, CA mark, final mark, grade, result code — all in one view.

```
CREATE VIEW Final_Marks_Summary AS
SELECT
    CU.Course_Code,
    CU.Course_Name,
    ST.Reg_No,
    P.F_Name,
    P.L_Name,
    ER.attendance_percentage,
    ER.ca_mark,
    FR.final_mark,
    FR.letter_grade,
    FR.grade_point,
    FR.result_code
FROM FINAL_RESULT FR
JOIN ELIGIBILITY_RECORD ER ON FR.Student_ID = ER.Student_ID AND FR.Offering_ID =
ER.Offering_ID
JOIN STUDENT ST ON FR.Student_ID = ST.Student_ID
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN COURSE_OFFERING CO ON FR.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID;
```

ELIGIBILITY & GRADE VIEWS created by Himansana (TG2096)

10. Student_Eligibility_Summary attendance status, CA mark, and both eligibility flags per student.

```
CREATE VIEW Student_Eligibility_Summary AS
SELECT
    CU.Course_Code,
    CU.Course_Name,
    ST.Reg_No,
    P.F_Name,
    P.L_Name,
    ER.attendance_percentage,
    CASE WHEN ER.attendance_percentage >= 80.00 THEN 'Eligible' ELSE 'Not Eligible' END
AS Attendance_Status,
    ER.ca_mark,
    ER.ca_eligibility,
    ER.final_exam_eligibility
FROM ELIGIBILITY_RECORD ER
JOIN STUDENT ST ON ER.Student_ID = ST.Student_ID
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN COURSE_OFFERING CO ON ER.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
ORDER BY CU.Course_Code, ST.Reg_No;
```

11. Student_Grades_Summary final mark, letter grade, grade point, result code per student.

```
CREATE VIEW Student_Grades_Summary AS
SELECT
    ST.Reg_No,
    CU.Course_Code,
    CU.Course_Name,
    CU.Credits,
    FR.final_mark,
    FR.letter_grade,
    FR.grade_point,
    FR.result_code
FROM FINAL_RESULT FR
JOIN STUDENT ST ON FR.Student_ID = ST.Student_ID
JOIN COURSE_OFFERING CO ON FR.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
ORDER BY ST.Reg_No, CU.Course_Code;
```

12. Batch_GPA_Summary SGPA and CGPA for all students, ordered by CGPA descending.

```
CREATE VIEW Batch_GPA_Summary AS
SELECT
    ST.Reg_No,
    P.F_Name,
    P.L_Name,
    SEM.Academic_Year,
    SEM.Semester_No,
    GR.sgpa,
    GR.cgpa
FROM GPA_RECORD GR
JOIN STUDENT ST ON GR.student_id = ST.Student_ID
JOIN PERSON P ON ST.Person_ID = P.Person_ID
JOIN SEMESTER SEM ON GR.semester_id = SEM.Semester_ID
ORDER BY GR.cgpa DESC;
```

13. Detailed_Academic_Transcript semester, course, grade, credits, SGPA, CGPA — full student transcript.

```
CREATE VIEW Detailed_Academic_Transcript AS
SELECT
    ST.Reg_No,
    SEM.Academic_Year,
    SEM.Semester_No,
    CU.Course_Code,
    CU.Course_Name,
    CU.Credits,
    FR.letter_grade,
    FR.grade_point,
```

```
    GR.sgpa,
    GR.cgpa
FROM FINAL_RESULT FR
JOIN STUDENT ST ON FR.Student_ID = ST.Student_ID
JOIN COURSE_OFFERING CO ON FR.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
JOIN SEMESTER SEM ON CO.Semester_ID = SEM.Semester_ID
JOIN GPA_RECORD GR ON ST.Student_ID = GR.student_id AND SEM.Semester_ID =
GR.semester_id
ORDER BY ST.Reg_No, CU.Course_Code;
```

OTHER VIEWS created by Nirmal (TG2097)

14. Student_Profile_View combined student personal data from PERSON and STUDENT.

```
CREATE OR REPLACE VIEW Student_Profile_View AS
SELECT
    S.Reg_no AS Reg_No,
    CONCAT(P.F_Name, ' ', P.L_Name) AS Full_Name,
    P.F_Name,
    P.L_Name,
    P.NIC,
    P.Gender,
    P.DOB,
    P.Phone_No,
    P.Address,
    S.Batch,
    S.Intake_Year AS Academic_Year,
    S.Status AS Student_Status
FROM STUDENT S
JOIN PERSON P ON S.Person_ID = P.Person_ID;
```

15. Low_Attendance_View students with attendance below 80%, worst first. Used by lecturers and admin.

```
CREATE OR REPLACE VIEW Low_Attendance_View AS
SELECT
    S.Reg_no AS Reg_No,
    CONCAT(P.F_Name, ' ', P.L_Name) AS Student_Name,
    S.Batch,
    CU.Course_Code,
    CU.Course_Name,
    ER.attendance_percentage,
    ER.final_exam_eligibility
FROM ELIGIBILITY_RECORD ER
JOIN STUDENT S ON ER.Student_ID = S.Student_ID
JOIN PERSON P ON S.Person_ID = P.Person_ID
JOIN COURSE_OFFERING CO ON ER.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
```

```
WHERE ER.attendance_percentage < 80.00
ORDER BY ER.attendance_percentage ASC;
```

16. Medical_Status_View medical submissions with course, session type, and approval status.

```
CREATE VIEW Medical_Status_View AS
SELECT
    M.Medical_ID,
    S.Reg_No,
    CONCAT(P.F_Name, ' ', P.L_Name) AS Student_Name,
    M.submitted_date,
    M.reason,
    CU.Course_Code,
    CC.component_type AS Session_Type,
    SE.week_no,
    SE.session_date AS Absent_Date,
    M.approval_status
FROM MEDICAL_RECORD M
JOIN STUDENT S ON M.Student_ID = S.Student_ID
JOIN PERSON P ON S.Person_ID = P.Person_ID
JOIN SESSION SE ON M.Session_ID = SE.Session_ID
JOIN COURSE_COMPONENT CC ON SE.Component_ID = CC.Component_ID
JOIN COURSE_OFFERING CO ON CC.offering_id = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID;
```

17. Session_Details_View session schedule with course name, component type, and department.

```
CREATE VIEW Session_Details_View AS
SELECT
    SE.Session_ID,
    CC.component_type AS Session_Type,
    SE.week_no,
    SE.session_date,
    SE.start_time,
    SE.end_time,
    CU.Course_Code,
    CU.Course_Name,
    CU.Credits,
    CU.Course_Type,
    D.Department_Name
FROM SESSION SE
JOIN COURSE_COMPONENT CC ON SE.Component_ID = CC.Component_ID
JOIN COURSE_OFFERING CO ON CC.offering_id = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
JOIN DEPARTMENT D ON CU.Department_ID = D.Department_ID;
```

18. Top_Performers_View students who scored 80 or above in any subject, ranked by final mark.

```sql
CREATE VIEW Top_Performers_View AS
SELECT
    S.Reg_No,
    CONCAT(P.F_Name, ' ', P.L_Name) AS Student_Name,
    S.Batch_Year AS Batch,
    CU.Course_Code,
    CU.Course_Name,
    FR.final_mark,
    FR.letter_grade AS Grade
FROM FINAL_RESULT FR
JOIN STUDENT S ON FR.Student_ID = S.Student_ID
JOIN PERSON P ON S.Person_ID = P.Person_ID
JOIN COURSE_OFFERING CO ON FR.Offering_ID = CO.Offering_ID
JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
WHERE FR.final_mark >= 80.00
ORDER BY FR.final_mark DESC;
```

6. Architecture of the Solution

The solution uses a two-layer architecture: a MySQL 8.x database server hosting the FOT_Academic_Management_DB, and an application/client layer that connects via the standard MySQL protocol.

6.1 Database Layer

- Engine: InnoDB (full ACID compliance, foreign key support, row-level locking)
- 24 relational tables organised in a 7-step dependency order
- 3 stored procedures for automated business logic (eligibility, results, GPA)
- 15+ database views providing safe, filtered read access
- Role-based MySQL user accounts with GRANT-level access control

6.2 Data Flow End to End

Step	Actor	Action
1	Admin	Creates PERSON, STAFF, STUDENT, USER_ACCOUNT records
2	Admin	Defines DEPARTMENT, SEMESTER, COURSE_UNIT records
3	Admin / Lecturer	Creates COURSE_OFFERING and assigns Lecturer; defines COURSE_COMPONENTs
4	Admin	Enrolls students: inserts ENROLLMENT rows (Student_ID, Offering_ID, attempt_no)
5	Lecturer	Defines ASSESSMENT_SCHEME (weight %, max_marks) per offering
6	Tech Officer	Creates SESSION rows; marks ATTENDANCE_RECORD per session per student
7	Student	Submits MEDICAL_RECORD linked to specific absent sessions
8	Lecturer	Enters STUDENT_MARK for each student per scheme
9	System (Procedure)	CALL Generate_Dynamic_Eligibility() — computes attendance_percentage and ca_mark
10	System (Procedure)	CALL Generate_Dynamic_Final_Result() — computes final_mark, assigns grade from GRADE_SCALE
11	System (Procedure)	CALL Generate_Dynamic_GPA() — computes credit-weighted SGPA and CGPA
12	Authorised Users	Query views (e.g. Detailed_Academic_Transcript, Final_Marks_Summary, Low_Attendance_View)

7. Tools and Technologies Used

Tool / Technology	Purpose
MySQL 8.x	Primary relational DBMS — tables, views, stored procedures, user accounts
MySQL Workbench	IDE for schema design, query execution, and user management
InnoDB Storage Engine	ACID-compliant transactions and foreign key enforcement
SQL (DDL / DML / DCL)	Table creation, data manipulation, and access control
Stored Procedures (PL/SQL)	Automated eligibility, final result, and GPA computation
Views	Read-only summarised access to derived data for different user roles
Git / GitHub	Version control with branches per module (Data_INSERT, Procedure, View, Dev)
draw.io	ER/EER and relational mapping diagram creation
Microsoft Word	Project documentation and reporting

8. Security Measures

8.1 Role-Based Access Control (RBAC)

All access to the database is controlled through MySQL user accounts. The principle of least privilege is strictly applied — no application or user connects as root.

User Role	Permissions Granted	Reason
Admin (admin_nalin)	ALL PRIVILEGES ON *.* WITH GRANT OPTION	Full system control including creating/revoking other user accounts
Dean (dean_saman)	ALL PRIVILEGES ON FOT_Academic_Management_DB.*	Needs full read/write for oversight, reporting, and approval workflows
Lecturer (lec_amali)	ALL PRIVILEGES ON FOT_Academic_Management_DB.*	Needs to enter marks, manage offerings, and view student data
Technical Officer (to_kamal)	SELECT, INSERT, UPDATE on ATTENDANCE_RECORD, SESSION; SELECT on attendance views only	Scope limited strictly to attendance management — no access to marks or results
Students (tg2095 etc.)	SELECT on Student_Profile_View, Student_Eligibility_Summary, Final_Marks_Summary, Student_Grades_Summary, Detailed_Academic_Transcript only	Read-only, through views only — cannot read raw tables or other students' data

8.2 Password Hashing

Passwords stored in the USER_ACCOUNT table should be hashed at the application layer (e.g. using bcrypt) before insertion. The VARCHAR(255) column accommodates standard hash lengths. Storing plain-text passwords in the database was used only during development and testing.

8.3 Referential Integrity

All 24 tables use InnoDB foreign keys. Orphaned records are prevented by the DBMS — for example, an ATTENDANCE_RECORD cannot reference a SESSION or STUDENT that does not exist, and deleting a DEPARTMENT would fail if students or staff reference it.

8.4 ENUM and NOT NULL Constraints

The Gender column in PERSON uses ENUM('Male','Female','Other') to restrict invalid values. Critical fields (F_Name, L_Name, NIC, hire_date, DOB, etc.) are NOT NULL. UNIQUE constraints on NIC, Reg_no, staff_no, and UserName prevent duplicate records.

8.5 View-Based Data Abstraction

Students never touch base tables. They only query predefined views that already filter data to the relevant scope. This prevents horizontal privilege escalation (one student reading another student's records).

9. Database Accounts / Users

The following MySQL user accounts were created and configured with role-appropriate permissions. This is the actual script from the MYSQL USERS file in the project repository.

Reasons for Creating These Accounts

Account	Reason
admin_nalin	System-wide administrator. Manages all user accounts, database structure, and has GRANT OPTION to delegate permissions.
dean_saman	Faculty Dean requires complete read/write access to all academic data for oversight, reporting, and final approvals.
lec_amali (+ other lecturers)	Lecturers need to enter marks for their offerings, manage assessment schemes, and view student results.
to_kamal (Technical Officer)	Manages lab sessions and records attendance. Restricted strictly to attendance tables and views — no access to marks or results.
tg2095–tg2113 (Students)	Can only view their own profile, eligibility, marks, grades, and transcript through predefined read-only views. No write access whatsoever.

10. Code Snippets Stored Procedures

Three stored procedures automate the core computational business logic. They must be called in order: Eligibility → Final Result → GPA.

10.1 Generate_Dynamic_Eligibility (TG2097 Nirmal)

This procedure clears and repopulates ELIGIBILITY_RECORD. For each enrollment, it computes attendance_percentage counting Present and Medical statuses ca_mark the weighted sum of CA-type assessments (Quiz, Project, Mid Theory, Mid Practical); and (c) the eligibility flags based on the 80% attendance threshold.

```
DROP PROCEDURE IF EXISTS Generate  Dynamic  Eligibility;

DELIMITER $$

CREATE PROCEDURE Generate  Dynamic  Eligibility()
BEGIN

  DELETE FROM ELIGIBILITY_RECORD;
  INSERT INTO ELIGIBILITY  RECORD (Eligibility  ID, Student  ID, Offering  ID, attendance  percentage,
ca  mark, ca  eligibility, final  exam  eligibility)
  SELECT

    CONCAT('ELI  ', E.Offering  ID, '  ', E.Student  ID),
    E.Student  ID,
    E.Offering  ID,

    COALESCE((
      SELECT (SUM(CASE WHEN A.attendance  status IN ('Present', 'Medical') THEN 1 ELSE 0 END) /
COUNT(*)) * 100
      FROM ATTENDANCE_RECORD A
      JOIN SESSION S ON A.session  id = S.Session  ID
      JOIN COURSE  COMPONENT C ON S.Component  ID = C.Component  ID
      WHERE A.student  id = E.Student  ID AND C.offering  id = E.Offering  ID
    ), 0) AS attendance_percentage,

    COALESCE((
      SELECT SUM(SM.raw_mark * (ASCH.weight_percentage / 100))
      FROM STUDENT_MARK SM
      JOIN ASSESSMENT  SCHEME ASCH ON SM.Scheme  ID = ASCH.Scheme  ID
      WHERE SM.Student  ID = E.Student  ID
        AND ASCH.Offering_ID = E.Offering_ID

        AND ASCH.Assessment  Type  ID IN ('AT  QZ', 'AT  PR', 'AT  MT', 'AT  MP')
    ), 0) AS ca  mark,
```

```
    FALSE,
    FALSE
  FROM ENROLLMENT E;

  UPDATE ELIGIBILITY_RECORD
  SET ca_eligibility = IF(attendance_percentage >= 80.00, TRUE, FALSE),
    final_exam_eligibility = IF(attendance_percentage >= 80.00, TRUE, FALSE);

END$$
DELIMITER ;
CALL Generate_Dynamic_Eligibility();
```

10.2 Generate_Dynamic_Final_Result (TG2096 Himansana)

This procedure clears and repopulates FINAL_RESULT. Students who are not eligible receive a final_mark of 0.00. Eligible students receive the full weighted sum of all assessment marks. Grades are then assigned by joining with GRADE_SCALE using FLOOR() to match integer ranges.

```
DELIMITER $$

CREATE PROCEDURE Generate_Dynamic_Final_Result()
BEGIN

  DELETE FROM FINAL_RESULT;


  INSERT INTO FINAL_RESULT (Result_ID, Student_ID, Offering_ID, final_mark, letter_grade, grade_point,
result_code, released_date)
  SELECT

    CONCAT('RES_', E.Offering_ID, '_', E.Student_ID) AS Result_ID,
    E.Student_ID,
    E.Offering_ID,


    CASE
      WHEN ELI.final_exam_eligibility = FALSE THEN 0.00
      ELSE COALESCE((
        SELECT SUM(SM.raw_mark * (ASCH.weight_percentage / 100))
        FROM STUDENT_MARK SM
        JOIN ASSESSMENT_SCHEME ASCH ON SM.Scheme_ID = ASCH.Scheme_ID
        WHERE SM.Student_ID = E.Student_ID AND ASCH.Offering_ID = E.Offering_ID
      ), 0)
    END AS final_mark,
```

```
    'TBD' AS letter_grade,
    0.00 AS grade_point,
    'TBD' AS result_code,
    CURDATE() AS released_date

  FROM ENROLLMENT E
  JOIN ELIGIBILITY_RECORD ELI ON E.Student_ID = ELI.Student_ID AND E.Offering_ID = ELI.Offering_ID;


  UPDATE FINAL_RESULT FR
  JOIN GRADE_SCALE GS ON FLOOR(FR.final_mark) BETWEEN GS.min_mark AND GS.max_mark
  SET
    FR.letter_grade = GS.letter_grade,
    FR.grade_point = GS.grade_point,

    FR.result_code = IF(FR.final_mark >= 35.00, 'PASS', 'FAIL');

END$$

DELIMITER ;


CALL Generate_Dynamic_Final_Result();
```

10.3 Generate_Dynamic_GPA (TG2095 Hasaranga)

This procedure clears and repopulates GPA_RECORD. For each student per semester, it computes the credit-weighted SGPA using the grade points from FINAL_RESULT and the credit values from COURSE_UNIT. As this is a single-semester system, CGPA equals SGPA.

```
DROP PROCEDURE IF EXISTS Generate_Dynamic_GPA;

DELIMITER $$

CREATE PROCEDURE Generate_Dynamic_GPA()
BEGIN

  DELETE FROM GPA_RECORD;


  INSERT INTO GPA_RECORD (gpa_id, student_id, semester_id, sgpa, cgpa)
  SELECT
    CONCAT('GPA_', CO.Semester_ID, '_', FR.Student_ID) AS gpa_id,
    FR.Student_ID AS student_id,
    CO.Semester_ID AS semester_id,
```

```
    CAST((SUM(CU.Credits * FR.grade  point) / SUM(CU.Credits)) AS DECIMAL(4,2)) AS sgpa,
    CAST((SUM(CU.Credits * FR.grade_point) / SUM(CU.Credits)) AS DECIMAL(4,2)) AS cgpa

  FROM FINAL  RESULT FR
  JOIN COURSE  OFFERING CO ON FR.Offering  ID = CO.Offering  ID
  JOIN COURSE_UNIT CU ON CO.Course_ID = CU.Course_ID
  GROUP BY FR.Student  ID, CO.Semester  ID;

END$$

DELIMITER ;


CALL Generate  Dynamic  Eligibility();
CALL Generate_Dynamic_Final_Result();
CALL Generate  Dynamic  GPA();


SELECT * FROM GPA_RECORD;
```

10.4 Additional Stored Procedures Report Generation & Data Retrieval

The following seven stored procedures provide parameterised report generation for the most common queries required by Admins, Lecturers, Technical Officers, and Students. Each procedure queries one of the views defined in Section 5.3, ensuring that the business logic remains centralised and consistent.

10.4 SP_Get_Students_By_Eligibility Eligible / Not Eligible Students for a Course (TG2095 -Hasaranga)

Returns all students in a given course who are either eligible or not eligible for the final exam. Accepts the course code and a boolean flag (TRUE = Eligible, FALSE = Not Eligible). Queries the Student_Eligibility_Summary view.

```
CREATE PROCEDURE SP_Get_Students_By_Eligibility(
    IN p_Course_Code VARCHAR(10),
    IN p_Is_Eligible BOOLEAN
)
BEGIN
    SELECT Reg_No, F_Name, L_Name, Course_Code, attendance_percentage, ca_mark,
        CASE WHEN final_exam_eligibility = 1 THEN 'Eligible' ELSE 'Not Eligible' END AS Eligibility
    FROM Student_Eligibility_Summary
    WHERE Course_Code = p_Course_Code AND final_exam_eligibility = p_Is_Eligible;
END$$
```

10.5 SP_Get_Student_Final_Marks Final Marks and Grades for a Student (All Subjects) (TG2095 - Hasaranga)

Returns the final mark, letter grade, and result code for all subjects of a specific student. Queries the Final_Marks_Summary view.

```
CREATE PROCEDURE SP_Get_Student_Final_Marks(
    IN p  Reg  No VARCHAR(20)
)
BEGIN
    SELECT Reg_No, Course_Code, Course_Name, final_mark, letter_grade, result_code
    FROM Final  Marks  Summary
    WHERE Reg_No = p_Reg_No;
END$$
```

10.6 SP_Get_One_Student_All_Attendance Attendance of a Student Across All Subjects (TG2096 – Himansana)

Returns the attendance percentage and eligibility status for a specific student across all enrolled subjects. Queries the Student_Eligibility_Summary view.

```
CREATE PROCEDURE SP_Get_One_Student_All_Attendance(
    IN p_Reg_No VARCHAR(20)
)
BEGIN
    SELECT Reg_No, Course_Code, Course_Name, attendance_percentage, Attendance_Status
    FROM Student_Eligibility_Summary
    WHERE Reg  No = p  Reg  No;
END$$
```

10.7 SP_Get_Student_CA_Course CA Mark of a Student for a Specific Course (TG2096 – Himansana)

Returns the CA mark and CA eligibility flag for one student in one specific course. Queries the Student_CA_Marks_Summary view.

```
CREATE PROCEDURE SP_Get_Student_CA_Course(
    IN p_Reg_No VARCHAR(20),
    IN p  Course  Code VARCHAR(10)
)
BEGIN
    SELECT Reg_No, Course_Code, Course_Name, ca_mark, ca_eligibility
    FROM Student  CA  Marks  Summary
    WHERE Reg_No = p_Reg_No AND Course_Code = p_Course_Code;
END$$
```

10.8 SP_Get_Batch_CA_Course CA Marks of the Entire Batch for a Specific Course (TG2097 – Nirmal)

Returns the CA mark and CA eligibility for every student in a given course. Queries the Batch_CA_Marks_Summary view. Useful for lecturers to review the batch before releasing results.

```
CREATE PROCEDURE SP_Get_Batch_CA_Course(
    IN p_Course_Code VARCHAR(10)
)
BEGIN
    SELECT Reg_No, F_Name, L_Name, Course_Code, ca_mark, ca_eligibility
    FROM Batch_CA_Marks_Summary
    WHERE Course_Code = p_Course_Code;
END$$
```

10.9 SP_Get_Batch_Attendance_Course Attendance of the Entire Batch for a Specific Course Student (TG2112 - Sarathchandra GSM)

Returns the attendance percentage and eligibility status for all students enrolled in a given course. Queries the Batch_Attendance_Summary view. Used by Technical Officers and Lecturers for attendance reporting.

```
CREATE PROCEDURE SP  Get  Batch  Attendance  Course(
    IN p  Course  Code VARCHAR(10)
)
BEGIN
    SELECT Reg  No, Course  Code, Attendance  Percentage, Eligibility  Status
    FROM Batch  Attendance  Summary
    WHERE Course  Code = p  Course  Code;
END$$
```

10.10 SP_Get_Student_Medicals All Medical Records Submitted by a Student (TG2112 - Sarathchandra GSM)

Returns all medical submissions for a specific student, including the absent session date, course code, reason, and approval status. Queries the Medical_Status_View.

```
CREATE PROCEDURE SP_Get_Student_Medicals(
   IN p_Reg_No VARCHAR(20)
)
BEGIN
   SELECT Medical_ID, Reg_No, Student_Name, submitted_date, reason, Course_Code, Absent_Date,
approval_status
   FROM Medical  Status  View
   WHERE Reg_No = p_Reg_No;
END$$
```

11. Problems Faced During Development

#	**Problem Encountered**
1	Foreign key dependency ordering: MySQL raises errors when creating a table that references another table not yet created. With 24 tables and complex cross-references, determining the correct creation order initially resulted in multiple FK violation errors.
2	The M:N relationship between STUDENT and COURSE_OFFERING was initially modelled without an associative entity, causing schema issues. It was later resolved by introducing the ENROLLMENT table with its own PK and attempt tracking.
3	CA mark computation in the stored procedure required filtering only CA-type assessments (AT_QZ, AT_PR, AT_MT, AT_MP). An initial version summed all assessments including finals, producing incorrect CA marks.
4	Attendance percentage initially only counted 'Present' status. The faculty policy of counting approved medical absences as attended was missed, leading to lower-than-expected attendance percentages and incorrect eligibility flags.
5	Git merge conflicts arose because four team members worked on separate branches (Data_INSERT, Procedure, View, Dev) and SQL files in the same folder were modified by multiple members simultaneously.
6	The GRADE_SCALE BETWEEN comparison on a DECIMAL final_mark column against INT range columns produced edge-case rounding errors (e.g. 84.99 not matching G_A's range of 75–84). The fix was wrapping final_mark with FLOOR() before the comparison.
7	MySQL's INT(11) deprecation warning appeared for Office_Phone in DEPARTMENT during testing on MySQL 8.0.17+. This was documented and flagged for migration to BIGINT in the production schema.
8	The DEAN table initially included an Office_Extension column that was present in the ER diagram but absent from the relational mapping. The inconsistency was resolved by aligning both diagrams in the final version.

12. Solutions to the Identified Problems

#	Solution Applied
1	A dependency-ordered table creation document (FULL_TABLE_CREATION) was produced listing all 7 steps from base tables to derived tables. Each team member ran the script in this order only.
2	The ENROLLMENT associative entity was added with its own Enrollment_ID PK, FKs to STUDENT and COURSE_OFFERING, and an attempt_no column to support repeat enrollments. 225 enrollment records (25 students × 9 offerings) were inserted.
3	Assessment Type IDs for CA were standardised as AT_QZ, AT_PR, AT_MT, AT_MP. The stored procedure was updated to filter ASCH.Assessment_Type_ID IN ('AT_QZ','AT_PR','AT_MT','AT_MP'), excluding finals.
4	The attendance query was updated to count WHERE attendance_status IN ('Present','Medical'), matching the faculty's official policy.
5	A branch-per-feature strategy was enforced. TG2095 (Hasaranga) was designated team lead and handled all merges to main after a team review. Merge conflicts were resolved manually before committing.
6	FLOOR(FR.final_mark) was applied before the BETWEEN comparison, ensuring that marks like 84.99 map correctly to the A grade band (75–84) without floating-point boundary issues.
7	Documented as a known issue for production migration. The column will be changed to BIGINT in the cloud-deployed version.
8	The FULL_TABLE_CREATION script was used as the single source of truth for all table definitions. Any discrepancies between individual member files were resolved against this master script.

13. Backend Hosting

13.1 Planned Hosting Platform: Amazon RDS for MySQL

The backend database is planned to be hosted on Amazon Web Services (AWS) using Amazon RDS for MySQL. During development and evaluation, the RDS Free Tier (db.t3.micro, 20 GB SSD, Single-AZ) will be used. For production, it can be scaled to Multi-AZ with automated failover.

13.2 Reasons for Selecting AWS RDS

- Managed service: AWS handles OS patching, database engine upgrades, automated daily snapshots, and point-in-time recovery automatically.
- MySQL compatibility: All existing DDL scripts, stored procedures, and views run on RDS for MySQL without modification.
- Scalability: Can scale instance size vertically or add read replicas horizontally as student population grows.
- Security: Supports VPC isolation, security groups (port 3306 whitelisting), SSL/TLS connections, and IAM database authentication.
- Multi-AZ: Automatic failover to a standby replica in a different availability zone ensures high availability for exam season.
- Free Tier: Suitable for academic project submission and testing at zero cost during the evaluation period.

13.3 Changes Required for Cloud Deployment

Change Required	Detail
User host binding	Change all 'user'@'localhost' to 'user'@'%' or restrict to the application server's specific IP/VPC CIDR range.
Password strength	Enable MySQL validate_password plugin via the RDS parameter group; enforce strong passwords for all user accounts.
SSL/TLS connections	Configure the MySQL client to require SSL. Download and reference the AWS RDS CA bundle (rds-ca-2019-root.pem) in connection strings.
Connection strings	Update all application connection strings to use the RDS endpoint hostname (e.g. fot-db.xxxx.rds.amazonaws.com) instead of 'localhost'.
RDS Parameter Group	Tune max_connections, innodb_buffer_pool_size, and slow_query_log based on expected concurrent user load.
Automated backups	Configure automated daily snapshots with 7-day retention and enable point-in-time recovery.
VPC & Security Groups	Restrict inbound port 3306 to the application server's security group only — never expose the RDS endpoint publicly.
INT(11) column migration	Change Office_Phone in DEPARTMENT from INT(11) to BIGINT to eliminate MySQL 8.x deprecation warnings in the production schema.

14. Individual Contribution to Backend Development

14.1 Contribution of Hasaranga K.H.M (Team Leader)

Index No: TG/2024/2095
Main Role: Project Coordination, Git Infrastructure Management, and Core Backend Architecture.

Overall Impact:

As the Team Leader, I established the project's technical framework and managed the collaborative development workflow using GitHub. I implemented a robust branching strategy, including dev for core development, Data_INSERT for population, and specialized branches for View and Procedure logic to ensure code integrity and conflict-free merging. My contribution focused on the system's structural normalization and the complex logic required for GPA and enrollment tracking.

Key Tasks Performed:

- Git Repository Management: Configured and maintained the central repository, managing merges and resolving version control conflicts across four feature branches.
- Relational Schema Design: Directed the translation of the ER/EER diagrams into a normalized 3NF relational database structure.
- Core Computational Logic: Developed the logic for GPA calculations and student enrollment tracking to ensure academic data accuracy.

Database Objects Created:

- **Tables:**
  - **ASSESSMENT_TYPE**: Defines various assessment categories like Quiz, Project, and Final Exams.
  - **GRADE_SCALE:** Stores the official mark-to-grade mapping and grade point values.
  - **COURSE_OFFERING:** Manages the specific instances of courses offered to batches by lecturers.
  - **COURSE_COMPONENT:** Breaks down offerings into Theory and Practical components.
  - **ENROLLMENT:** An associative entity tracking student attempts and status for course offerings.
  - **STUDENT_MARK:** Records individual marks for each student per assessment scheme.
  - **ELIGIBILITY_RECORD:** Stores computed attendance percentages and CA marks for exam qualification.
  - **FINAL_RESULT:** Contains the final weighted marks and released grades for each course.
  - **GPA_RECORD:** Stores semester-wise SGPA and cumulative CGPA per student.
  - **ASSESSMENT_SCHEME:** Defines the weightage and mandatory status for various assessments.

- **Views:**
  - **Batch_Attendance_Summary:** Provides an overview of attendance and eligibility for an entire batch.
  - **Component_Wise_Attendance:** Displays attendance percentages split between Theory and Practical sessions.
  - **Individual_Attendance_Summary:** Summarizes attendance percentage for a specific student across subjects.
  - **Individual_Subject_Attendance_Detail:** Shows session-by-session attendance history for individual students.

- **Stored Procedures:**
  - **Generate_Dynamic_GPA:** Automatically computes credit-weighted SGPA and CGPA for all students.
  - **SP_Get_Students_By_Eligibility:** Retrieves lists of students eligible or ineligible for exams in a specific course.
  - **SP_Get_Student_Final_Marks:** Generates a report of final marks and grades for a specific student registration number.

GitHub Link: [https://github.com/Hasaranga-kariyawasam]

14.2 Contribution of Himansana N.V.S

Index No: TG/2024/2096
Main Role: Result Generation Logic and Role-Based Access Control (RBAC).

Overall Impact:

Himansana was responsible for the critical automation of final results and the security layer of the database. By developing advanced result-calculation procedures and configuring role-based user accounts, he ensured the system is both functional and secure from unauthorized access.

Key Tasks Performed:

- **Security & Access Control:** Authored the MYSQL USERS script to implement the principle of least privilege for different user roles.
- **Automated Result Computation:** Developed stored procedures to process raw marks into final grades using floor-rounded comparisons.
- **Data Initialization:** Managed the population of primary staff, department, and semester entities.

Database Objects Created:

- **Views:**
  - **Student_Eligibility_Summary:** Shows attendance status, CA marks, and eligibility flags per student.
  - **Student_Grades_Summary:** Provides a summary of final marks, grades, and result codes for each student.
  - **Batch_GPA_Summary**: Ranks the entire batch based on CGPA for academic performance review.
  - **Detailed_Academic_Transcript:** Generates a full transcript including grades, credits, and GPA records.
- **Stored Procedures:**
  - **Generate_Dynamic_Final_Result:** Computes total weighted marks and assigns letter grades based on eligibility.
  - **SP_Get_One_Student_All_Attendance:** Reports a specific student's attendance percentage and eligibility status across all subjects.
  - **SP_Get_Student_CA_Course:** Retrieves the CA mark and eligibility status for a student in a particular course.

GitHub Link: [https://github.com/himansana2004]

14.3 Contribution of Nirmal I.D.N.S

Index No: TG/2024/2097
Main Role: User Hierarchy Management and Eligibility Computation.

Overall Impact:
Nirmal focused on the fundamental identity management of the system, designing the generalization hierarchy for diverse personnel. He implemented the essential logic that determines exam eligibility by integrating medical and session attendance data, ensuring institutional policies are met.

Key Tasks Performed:
- **User Specialization Modeling:** Designed and created the tables for specialized staff roles including Admin, Dean, and Technical Officer.
- **Eligibility Logic Implementation:** Developed procedures to calculate qualification status based on attendance thresholds.
- **Comprehensive Mark Entry:** Managed the bulk insertion of student mark data to test reporting views.

Database Objects Created:
- **Tables:**
  PERSON, STAFF, STUDENT, USER_ACCOUNT: Foundation tables for managing personal and login data.
  ADMIN, DEAN, LECTURER, TECHNICAL_OFFICER: Specialized tables for different university personnel roles.
- **Views:**
  - **Student_Profile_View:** Combines personal details with academic registration information.
  - **Low_Attendance_View:** Identifies students with attendance below 80% for academic intervention.
  - **Medical_Status_View:** Tracks submitted medical records and their respective approval statuses.
  - **Session_Details_View:** Provides a comprehensive schedule including course details and locations.
  - **Top_Performers_View**: Lists high-achieving students who scored 80 or above in any subject.
- **Stord Procedures:**
  - **Generate_Dynamic_Eligibility:** Automatically clears and computes student CA marks and final exam eligibility.
  - **SP_Get_Batch_CA_Course:** Provides a report of CA marks and eligibility for all students in a given batch.

GitHub Link: [https://github.com/nirmalsasindu40]

14.4 Contribution of Sarathchandra G.M.S

Index No: TG/2024/2112
Main Role: Academic Core Infrastructure and Marks Analysis.

Overall Impact:

Sarathchandra was responsible for building the core academic components of the database, from department definitions to session records. He developed advanced marks views that allow for multi-dimensional analysis of student performance, facilitating efficient reporting for lecturers and admins.

Key Tasks Performed:

- **Infrastructure Setup:** Created the baseline tables for university departments, semesters, and individual course units.
- **Attendance Tracking Logic:** Implemented the storage for sessions and attendance records to support eligibility flows.
- **Data Integration:** Managed the population of 225 enrollment records and session scheduling data.

Database Objects Created:

- **Tables:**
  DEPARTMENT, SEMESTER, COURSE_UNIT: Essential academic organizational tables.
  SESSION, ATTENDANCE_RECORD, MEDICAL_RECORD: Core tables for tracking daily academic activity and absences.
- **Views:**
  - **Individual_CA_Marks_Detail:** Provides a detailed weighted breakdown of all CA components per subject.
  - **Student_CA_Marks_Summary:** Displays summarized CA results and eligibility for students.
  - **Batch_CA_Marks_Summary:** Summarizes CA marks for an entire batch for lecturer review.
  - **Comprehensive_Marks_Breakdown:** Pivots all assessment types into columns for a side-by-side comparison.
  - **Final_Marks_Summary:** A complete academic record view combining attendance, CA, and final results.
- **Stored Procedures:**
  - **SP_Get_Batch_Attendance_Course:** Retrieves the attendance percentage and eligibility for a whole batch in a specific course.
  - **SP_Get_Student_Medicals:** Lists all medical submissions for a student including reasons and approval status.

GitHub Link: [https://github.com/samindi2004]

15. References

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GitHub Repo Link: [https://github.com/Hasaranga-kariyawasam/DBMS_MINI_PROJECT]